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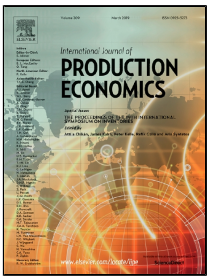
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Effects of design for the environment on firms' production and remanufacturing strategies

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Abstract: Firms have increasingly adopted the design for the environment (DfE) strategy because environmental regulations are gradually becoming more stringent. This can make products more recyclable and environmentally-friendly. In this study, we consider two cases, namely, a monopolistic case in which the original equipment manufacturer (OEM) and an independent remanufacturer (IR) are integrated and a competitive case in which OEM and IR are independent. A theoretical model is proposed to explore the impacts of DfE on firms' quantity decisions, as well as the impacts of competition on the DfE decision. Comparisons of the optimal decisions in the two scenarios yield some interesting results. In the monopolistic case, DfE reduces the OEMs' willingness to engage in remanufacturing, but the outcome is completely reversed in the competitive case in which DfE will incentivize IR to enter the market and cannibalize OEM's market share. Second, the optimal DfE effort level can be higher (or lower) in the competitive case than that in the monopolistic case, which relies on the relationship between the scale parameter of the investment cost of DfE and the profitability of remanufacturing. In addition, we find that under some conditions a high level of DfE may hurt the environment through the greatly increased total sales, although DfE can help reduce the environmental impact of unit product.

Keywords: Design for the environment; Remanufacturing; Competition; Cannibalization problem

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1 Introduction

As an effective means to reduce resource waste and production costs, remanufacturing has elicited extensive attention from the industry and academia. Improving consumers' environmental awareness encourages original equipment manufacturers (OEMs) to engage in reverse logistics management and recover and recycle their used products through remanufacturing Wu (2012). In 2016, Canon extracted 19,350 tons of plastics from old products and used them as raw materials for new products, and the company also recovered 14,853 tons of products and components[1]. To promote sustainable development and remanufacturing efficiency, Xerox established several international recycling systems. In January 2008, Xerox set up a recycling system in the Chinese Mainland to collect old products and ink cartridges. In 2012, Xerox expanded the recycling system to Korea, Australia, and New Zealand[2].

However, firms have increasingly begun to incorporate remanufacturing into their product design process because regulations on the environment and resources have gradually become more stringent. Through strategic product design and by improving the remanufacturing technology, products become environmentally friendly and easy to be remanufactured. For example, Canon considers recovery efficiency and recyclability in the product design phase and uses recyclable and reusable materials during the production process. Old products are then recycled through the Product-to-Product Recycling Program. Hewlett-Packard considers the recyclability of products in the product design stage[3]. Furthermore, in the automobile industry, many firms are starting to invest in the development of hybrid vehicles and new energy vehicle, such as Toyota and BYD.

These actions implemented by Canon and Hewlett-Packard make up a strategy called *design for the environment* (DfE) (Chen, 2001; Raz et al., 2013). The term DfE has many similar expressions according to where it is used. In Europe, it is primarily described as ecological design or life-cycle design. In addition, the DfE can be extended to sustainable product design or design for sustainability

[1] Canon sustainability report 2017. Available at: https://www.usa.canon.com/internet/portal/us/home/about/environment-sustainability-initiatives.
[2] Xerox sustainability report 2017. Available at: http://www.fujifilmholdings.com/en/sustainability/report/download/report2017.html.
[3] Hewlett-Packard Sustainability Report 2016. Available at: http://www8.hp.com/us/en/hp-information/global-citizenship/index.html#.

by considering the social and economic issues. More details of the term DfE can be found in Kurk and Eagan (2008). During the implementation of DfE, firms consider resource efficiency, environmental impact and remanufacturing during the design of products. Firms can reduce resource consumption and production costs through DfE. For example, with the improvement of technologies for collection and product design, Canon has reduced its reliance on natural resources, waste and operation costs, and eventually provided a stable supply of resources.

In particular, Figure 1 describes the practice of DfE in Canon Inc[4]. In the product planning and development stage, Canon invests in DfE, and develops some environmentally-conscious products in terms of resource saving, energy saving and recycling. Then, Canon carries out green procurement and manufacturing, which helps reduce waste generation and resource consumption. Next, products are sold and delivered to end consumers in the market. Finally, Canon recycles the used products from consumers, and reuses, refurbishes, and remanufactures them.

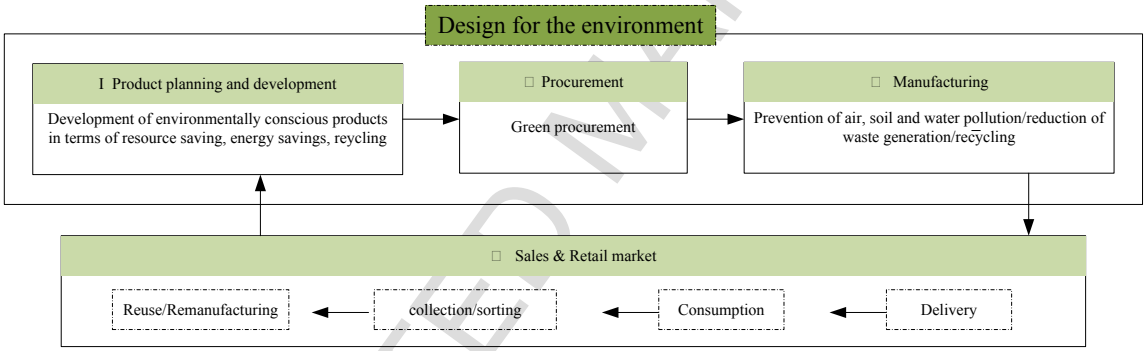


Figure 1. Practice of DfE in Canon Inc.

Obviously, the implementation of DfE is helpful to reduce resource consumption and waste and improve the product's reusability although firms have to invest a lot of money in DfE. Therefore, a challenge of DfE is the strategic balance between the decrease of production and remanufacturing costs and the huge investment cost. Furthermore, DfE can benefit remanufacturing because of products becoming more recyclable and easier to remanufacture. Therefore, used products may bring high economic value since DfE can also reduce the remanufacturing cost. In this case, it is intuitive that OEMs will become more likely to undertake remanufacturing as long as they operate

[4] For more details, please refer to http://www.canon.com/csr/.

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remanufacturing and manufacturing together. However, in such a situation where manufacturing is operated by OEMs while remanufacturing is undertaken by some independent remanufacturers (IRs), IRs may be encouraged to enter the market and cannibalize the OEMs market share with the presence of DfE. Consequently, DfE may further aggravate the competition between OEM and IR. In addition, since DfE can reduce resource consumption and production costs, it may improve the total sales of products. Thus, the effects of DfE on the overall environmental impact are unclear although DfE can reduce the environmental impact of unit product.

Motivated by this, we seek to resolve several important questions when factoring in the presence of DfE. To be specific, our study mainly seeks to answer the following questions.

1. Given that DfE is beneficial in reducing production and remanufacturing costs, how does it influence a firm's willingness to undertake remanufacturing in monopolistic and competitive circumstances?
2. Given that DfE helps reduce the environmental impact of unit products, does it also help increase the total environmental impact in monopolistic and competitive circumstances?
3. What are the effects of competition on the firm's DfE strategy compared to the monopolistic case?

In this study, we build a theoretical model that captures an essential characteristic of DfE, i.e., DfE can reduce production and remanufacturing costs. Based on this statement, we consider two cases, namely, a monopolistic case in which the OEM and IR are integrated and a competitive case in which the OEM and IR act independently. We further explore how DfE affects OEM's operations and environmental impact. In addition, we study the impacts of the competition between the OEM and the IR on the OEM's optimal DfE strategy.

The rest of the paper is organized as follows. Section 2 presents a literature review, and Section 3 provides related notations and assumptions. The results are derived and analyzed in Section 4, and an optimal strategy of DfE is investigated in Section 5. The conclusions of this study are presented in Section 6.

2 Literature review

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Two literature streams are closely related to our study, including: (1) Sustainable operations management and (2) sustainability study of DfE.

2.1 Sustainable operations management

Scholars have conducted in-depth research on sustainable management from various perspectives but focused on pricing and remanufacturing strategies (Atasu et al., 2008; Debo et al., 2005; Gan et al., 2017; Liu et al., 2018), reverse channel management (Choi et al., 2013; Savaskan et al., 2004), interaction with environmental regulations (Atasu and Subramanian, 2012), and green supply chains (Fahimnia et al., 2015a; Govindan et al., 2014; Surajit et al., 2017). However, the most closely related topic to our study is pricing and remanufacturing strategies. Several scholars have studied this issue in monopolistic environments. Ferrer and Swaminathan (2006) studied the pricing and remanufacturing strategies of a monopoly under the assumption that new and remanufactured products are homogeneous. On the basis of this argument, Ferrer and Swaminathan (2010) further studied this issue by considering that new and remanufactured products are heterogeneous. Chen and Chang. (2013) analyzed the dynamic pricing of new and remanufactured products under the consideration of product differentiation. Xiong et al. (2013) discussed the impact of remanufacturing strategies on suppliers. Through empirical research, Abbey et al. (2015) discovered that when new and remanufactured products are sold simultaneously, companies should increase the price of new products rather than reduce it. Aydin et al. (2015) proposed a novel approach to solve the firms' strategy of product line design when taking into account new and remanufactured products together. Wu et al. (2016) studied the joint decisions of product design and remanufacturing to minimize the total production cost and maximize the total market share. Yan et al. (2017) analyzed a firm's optimal pricing and manufacturing strategies when remanufactured products are sold together with new products. They showed that the firm' manufacturing strategies (make-to-order and make-to-store) had important effects on its' pricing strategies. Cui et al. (2017) studied the choice of remanufacturing quality strategy while considering consumer preferences. Steeneck and Sarin (2018) explored the effects of extended producer responsibility on product design when an OEM undertakes remanufacturing and leasing. Chai et al. (2018) examined whether carbon caps and trade mechanisms

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are beneficial in remanufacturing or not. Miao et al. (2018) investigated the effects of carbon regulation on firms' remanufacturing and trade-in policies. Different from the works above, we consider not only the monopolistic case, but also the competitive situation. In particular, we study the pricing and remanufacturing strategies when considering the impacts of DfE.

Other scholars have studied pricing and remanufacturing strategies in competitive environments. Ferrer and Swaminathan (2010) examined quantity competition between OEMs and IRs in consideration of product heterogeneity. Wu (2013), who considered the competition among duopoly equipment manufacturers, found that the entry of IRs may be beneficial to OEMs. Örsdemir et al. (2014) modeled the optimization of quality selection and remanufacturing strategies of OEMs in a competitive environment. To explore tradeoffs between cost and environmental degradation, energy consumption and waste generation, Fahimnia et al. (2015b) proposed a tactical supply chain planning model. Li et al. (2016) studied the optimal pricing strategy in a competitive dual-green supply chain. Wu and Zhou (2016) determined if the entry of a third-party remanufacturer negatively affects OEMs. The current study is closest to the study of Wu (2013). With the Hotelling model, Wu (2013) studied price competition and product interchangeability design in a competitive environment. In Wu's study, all consumers in the market can purchase products, and interchangeability design may lead to an increased production cost. However, we focus on quantity competition between OEMs and IRs, and consumers can independently decide whether to buy products or not. In addition, DfE can help reduce the production costs of new and remanufactured products.

2.2 Sustainability study of DfE

Several studies have also been conducted on sustainability of DfE. To some extent, DfE is similar to product design and product greenness design (Raz et al., 2013). Details of the definition of DfE can be found in Kurk and Eagan (2008). Based on Atasu et al. (2008), Wu (2012) considered product design and pricing strategy and explored a firm's choice of high and low disassemblability by constructing a two-period model. Atasu and Souza (2011) studied the issue of product quality and remanufacturing in consideration of government recycling regulations. Raja Ghazilla et al. (2015) investigated the existing state of the practice of DfE and design for disassembly in Malaysia. From the

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viewpoint of DfE, Gouda et al. (2016) studied the impacts of regulatory policies on firms' product development strategies. Jackson et al. (2016) examined the relationship of DfE and quality management innovation to environmental and economic performance. This work extended the study on the interaction mechanism between quality management and sustainable operations management. Sousa-Zomer et al. (2018) explored how clean production principles and practices adopted by companies in an emerging economy help develop a circular economy. Sonego et al. (2018) investigated the relationship between modularity and sustainable design on the basis of product life cycle.

The most relevant studies are those of Chen (2001) and Raz et al. (2013). With the quality-based model, Chen (2001) studied the mechanism of consumer preference, product strategy, and government environmental protection regulations. The results showed that the development of green products and strict environmental protection regulations do not necessarily improve environmental impacts. Using the newsvendor model, Raz et al. (2013) studied a firm's optimal quantity and DfE strategies and analyzed the influence of demand uncertainty on the firm's decisions. The research presented above focused on the optimization problems of companies in a monopolistic context. However, our study focuses on the impact of DfE on remanufacturing strategy and environmental impact and explores the role of competition in a firm's DfE strategy.

2.3 Summary

Our study contributes to the literature as follows: first, we focus on studying how the firms' DfE strategy affects their (or their competitors') willingness for remanufacturing under different (i.e., monopolistic or competitive) circumstances. This is completely different from the existing studies on sustainable operations management, which focused on exploring firms' remanufacturing strategies when considering the impact of DfE. Second, we focus on examining how horizontal competition affects the firms' optimal DfE and remanufacturing strategies. This is also distinct from current works on DfE which generally study the impacts of DfE on firms' manufacturing and remanufacturing strategy. Based on the theoretical analyses and numerical examples, we found some interesting and meaningful results which provide some helpful guidelines for firms' practical operations in DfE.

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3 Notations and assumptions

We consider an OEM who sells new products and an IR who sells remanufactured products. In the first stage, the OEM decides the DfE effort level (e). In the second stage, the OEM determines the quantity of new products (q_n), and the IR determines the quantity of remanufactured products (q_r). We assume that both the new and remanufactured products are sold in a single period, and the OEM and the IR simultaneously determine the quantity decisions. In other words, a Nash game on quantity competition occurs. In this paper, we mainly study the steady-state equilibrium, which enables us to obtain tractable analytical results and focus on the essential research problems. Similar assumption can be seen in Atasu and Souza (2011) and Örsdemir et al. (2014).

Table 1 summarizes the relevant notations used in the paper.

Table 1. Summary of related notations

Symbol	Definition
v	Consumers' willingness to pay for new products
ϕv	Consumers' willingness to pay for remanufactured products, $0 \le \phi < 1$ is a discount factor
c_n	Baseline production cost of unit of new product
c_r	Baseline production cost of unit of remanufactured product
β_1	Marginal effect of DfE on new products' production cost, $\beta_1 > 0$
β_2	Marginal effect of DfE on remanufactured products' production cost, $\beta_2 > 0$
$\tilde{c}_n$	Production cost of unit of new (or remanufactured) product with DfE
$\tilde{c}_r$	Production cost of unit of remanufactured product with DfE
δ_n	Baseline environmental effect of unit of new product, $\delta_n > 0$
δ_r	Baseline environmental effect of unit of remanufactured product, $\delta_r > 0$
ρ_1	Marginal effect of DfE on new products' environmental impact
ρ_2	Marginal effect of DfE on remanufactured products' environmental impact
$\tilde{\delta}_n$	Effect of new products on the environment with DfE
$\tilde{\delta}_r$	Effect of remanufactured products on the environment with DfE
F	Investment function of DfE
k	Scale parameter of the investment function
π_i^j	Profit function of firm i, where $i = \{I,OEM,IR\}$, I represents the integrated firm, OEM and IR respectively denote the OEM and IR, $j = \{M,C\}$, and M and C respectively denote the monopolistic and competitive cases
Decision variables	
$q_n(q_r)$	Quantity of new (or remanufactured) products
e	Effort level of DfE, and $0 \le e \le 1$

Next, we list some important assumptions related to our study, and provide explanations.

Assumption 1. The baseline production cost of new products is higher than the baseline remanufacturing cost, i.e., $c_n > c_r$.

This assumption is in accordance with that of Xiong et al. (2013) and Ferguson and Toktay (2006). In fact, Giutini and Gaudette (2003) showed that the unit remanufacturing cost approximately

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accounts for 40%–65% of that of a new product.

Assumption 2. Consumers' willingness to pay for a new product unit is heterogeneous and uniformly distributed over $[0,1]$, i.e., $v \sim U(0,1)$, while consumers' willingness to pay for a remanufactured product unit is a fraction of that of a new product, i.e., ϕv and $0 \le \phi < 1$. Without loss of generality, the market size is normalized to 1.

In reality, consumers prefer to pay a higher price for new products than remanufactured products, because differentiation in consumers' perceived valuation of new and remanufactured products always exists even though they have the same functions. Assumption 2 is in accordance to related works on operations management and marketing, such as Ferguson and Toktay (2006), Xiong et al. (2013) and Örsdemir et al. (2014).

Assume that consumer's surplus from purchasing a unit of new product is $v - p_n$, then the surplus from buying a unit of remanufactured product is $\phi v - p_r$ on the basis of Assumption 2. Clearly, a consumer prefers to buy a new product if $v - p_n > \phi v - p_r$ and $v - p_n \ge 0$, or a remanufactured product if $\phi v - p_r > v - p_n$ and $\phi v - p_r \ge 0$. Based on these analyses, the prices for new and remanufactured products can be respectively expressed as

$$p_n = 1 - q_n - \phi q_r, \tag{1}$$

$$p_r = \phi(1 - q_n - q_r). \tag{2}$$

Assumption 3. DfE can help reduce the marginal production costs of new and remanufactured products. In particular, the cost functions are inversely proportional to DfE effort level e: $\tilde{c}_n(e) = (1 - \beta_1 e)c_n$ and $\tilde{c}_r(e) = (1 - \beta_2 e)c_r$, $\beta_1, \beta_2 > 0$ and $\beta_1 \ne \beta_2$.

This Assumption captures the practices that firms can reduce the consumption of resources through investment in DfE. Canon has reduced its reliance on natural resources through investment in DfE. Meanwhile, Xerox and Hewlett-Packard have also reduced their production costs by adopting policies similar to DfE. Furthermore, similar phenomena can be found in the Automobile and printer industries. Many firms invest in hybrid vehicles and new energy vehicles to meet the governmental requirements on reducing the consumption of natural resources. A typical example of this is the development of BYD. For BYD, with the investment of the R&D of battery, BYD reduces the

production cost of new energy vehicles (BYD CSR Report 2017). According to some reports, the investment of the R&D of new energy vehicles minimizes the production cost up to 30%[5]. In addition, Toyota also put its efforts into the development of hybrid vehicles. According to some news, the investments in hybrid vehicle help Toyota reduce its production cost significantly, which is up to 1/3[6]. In addition, Assumption 3 aligns with the works of Chen (2001), Raz et al. (2013) and Wu (2013). In the study of Wu (2013), he assumed firms would reduce the production costs of new and remanufactured products when the products become more exchangeable. If $\beta_1 = \beta_2$, we can prove that the conditions where firms undertake "no remanufacturing," "partial remanufacturing," and "all remanufacturing" sequentially are the same as those in the absence of DfE. Thus, we consider two cases in our study, i.e., $\beta_1 > \beta_2$ and $\beta_1 < \beta_2$.

Assumption 4. The investment in DfE ($F(e)$) is quadratic and satisfies $dF(e)/de > 0$, $d^2F(e)/de^2 > 0$ and $F(0) = 0$. Specifically, we assume that it is $F(e) = ke^2/2$, where $k > 0$ is the scale coefficient.

This assumption coincides with the law of diminishing return to investment. Similar forms of response functions have been widely used in advertising response models of consumer retention and product awareness and in salesforce effort response models in marketing. In fact, Assumption 4 aligns with the work of Wu (2012), Raz et al. (2013), and Wu (2013).

Assumption 5. Parameter k satisfies $k > k_{\min} = \frac{\phi c_n^2 \beta_1^2 - 2\phi c_n c_r \beta_1 \beta_2 + c_r^2 \beta_2^2}{2(1-\phi)\phi}$.

Assumption 6 illustrates that DfE is costly. Hence, sparing no effort to do so is not economically viable (see similar assumptions in Savaskan et al. (2004)). In addition, this assumption ensures the objective functions are concave in DfE effort level in the first stage.

Assumption 6. The total effects of new and remanufactured products on the environment are $\tilde{\delta}_n$ and $\tilde{\delta}_r$, respectively, which are inversely proportional to DfE effort level e, that is, $\tilde{\delta}_n(e) = (1 - \rho_1 e)\delta_n$, $\tilde{\delta}_r(e) = (1 - \rho_2 e)\delta_r$, and $\rho_1, \rho_2 > 0$. In addition, $\tilde{\delta}_n(0) \geq \tilde{\delta}_r(0)$, i.e., $\delta_n > \delta_r > 0$.

In reality, firms could utilize various measures for different impact categories based on Life

[5] www.byd.com.cn; http://www.sohu.com/a/86924765_183181.
[6] https://www.cnbeta.com/articles/tech/751193.htm; http://baijiahao.baidu.com/s?id=1603340140849447410&wfr=spider&for=pc.

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Cycle Assessment methodology. For simplicity, we use a notation to represent the environmental impact of a unit of product which relies on the baseline environmental impact and DfE effort level. Reusable materials and components can be utilized during remanufacturing, and DfE can reduce material usage. Therefore, the environmental impact of a unit of product decreases by investing in DfE.

For example, the environment-oriented design of Huawei plays a significant role in reducing environmental impact. In 2017, the reuse design of Huawei's products achieved a high level of reuse rate of 81.2%, which improved the utilization rate of network equipment products. In addition, in the 5-year period of the next-generation access 2.0 project, Huawei achieved the goal of reducing CO2 emissions by an average of 11% and reducing energy consumption by 8%, helping to achieve BT energy-saving and emission reduction targets[7]. In fact, Assumption 7 is in accordance with the studies of Thomas (2011), Agrawal et al. (2012), Xiong et al. (2013), and Zhang and Zhang (2018). In addition, Assumption 7 implies that the environmental effects of new and remanufactured products with DfE depend on the DfE effort and the corresponding baseline environmental effects.

According to Agrawal et al. (2012), Zhang and Zhang (2018) and Xue et al. (2018), the total environmental impact can be obtained as

$$E = \tilde{\delta}_n q_n + \tilde{\delta}_r q_r = (1 - \rho_1 e)\delta_n q_n + (1 - \rho_2 e)\delta_r q_r. \quad (3)$$

Note that the higher the value of E, the worse the environmental impact.

4 Model and results

In this section, we consider two cases, i.e., a monopolistic case in which the OEM and the IR are integrated and a competitive case in which the OEM and the IR are independent and compete in terms of quantity. To study how the DfE policy affects firms' production and remanufacturing strategies, we study the optimization problems by considering an exogenous DfE effort level.

4.1 Quantity decisions in a monopolistic case

Given that OEM and IR are assumed to be integrated in the monopolistic situation, the OEM

[7]https://www-file.huawei.com/-/media/corporate/pdf/sustainability/2017-huawei-sustainability-report-cn.pdf?la=zh&source=corp_comm

decides the quantities of new and remanufactured products simultaneously given the DfE effort level. According to Equations (1) and (2), we obtain the profit-maximization problem as follows:

$$\pi_I^M(q_n,q_r|e) = q_n(p_n - \tilde{c}_n(e)) + q_r(p_r - \tilde{c}_r(e)) - F(e), \quad (4)$$

$$\text{s.t. } q_n \geq q_r \geq 0.$$

Given DfE effort level e, we can simply prove that $\pi_I^M(q_n,q_r|e)$ is jointly concave in q_n and q_r. Therefore, we obtain the optimal quantities, which are given by Lemma 1. *Note that all proofs of the results are provided in Section I in the Appendix.*

Lemma 1. Given DfE effort level e, the integrated firm's optimal quantity decisions are as follows:

(1) If $\frac{1-e\beta_1}{1-e\beta_2} \leq \frac{c_r}{\phi c_n}$, then $q_r^{M*} = 0$ and $q_n^{M*} = \frac{1-c_n(1-e\beta_1)}{2}$.

(2) If $\frac{c_r}{\phi c_n} < \frac{1-e\beta_1}{1-e\beta_2} < \frac{1-\phi}{2c_n(1-e\beta_2)} + \frac{(1+\phi)c_r}{2\phi c_n}$, then

$$q_r^{M*} = \frac{\phi c_n(1-e\beta_1) - c_r(1-e\beta_2)}{2(1-\phi)\phi} \text{ and } q_n^{M*} = \frac{1-\phi-c_n(1-e\beta_1)+c_r(1-e\beta_2)}{2(1-\phi)}.$$

(3) If $\frac{1-e\beta_1}{1-e\beta_2} \geq \frac{1-\phi}{2c_n(1-e\beta_2)} + \frac{(1+\phi)c_r}{2\phi c_n}$, then $q_r^{M*} = q_n^{M*} = \frac{1+\phi-c_n(1-e\beta_1)-c_r(1-e\beta_2)}{2+6\phi}$.

Consequently, the corresponding profits are summarized in Table 2.

Table 2. The integrated firm's profit in the monopolistic case

Conditions	$\pi_I^M(e)$
$\frac{1-e\beta_1}{1-e\beta_2} \leq \frac{c_r}{\phi c_n}$	$\frac{1-c_n(1-e\beta_1)(2-c_n(1-e\beta_1))}{4} - \frac{e^2k}{2}$
$\frac{c_r}{\phi c_n} < \frac{1-e\beta_1}{1-e\beta_2} < \frac{1-\phi}{2c_n(1-e\beta_2)} + \frac{(1+\phi)c_r}{2\phi c_n}$	$\frac{(1-\phi)\phi + \phi c_n^2(1-e\beta_1)^2 + c_r^2(1-e\beta_2)^2 - 2\phi c_n(1-e\beta_1)(1-\phi+c_r(1-e\beta_2))}{4(1-\phi)\phi} - \frac{e^2k}{2}$
$\frac{1-e\beta_1}{1-e\beta_2} \geq \frac{1-\phi}{2c_n(1-e\beta_2)} + \frac{(1+\phi)c_r}{2\phi c_n}$	$\frac{(1+\phi-c_n(1-e\beta_1)-c_r(1-e\beta_2))^2}{4(1+3\phi)} - \frac{e^2k}{2}$

Lemma 1 indicates that the firm can select different production and remanufacturing policies depending on the DfE effort level. DfE reduces the production costs of new and remanufactured products, which further intensifies the competition between the two products. With the increase of the DfE effort level, i.e., $(1-e\beta_1)/(1-e\beta_2)$ decreases, the firm selects "all remanufacturing," "partial remanufacturing," and "no remanufacturing" strategies in turn.

Based on Lemma 1, we obtain the following result.

Proposition 1. In comparing the scenario without DfE (i.e., $e = 0$), with the presence of DfE, the

integrated firm becomes less willing to undertake remanufacturing if $\beta_1 > \beta_2$, and becomes more willing to undertake remanufacturing if $\beta_1 < \beta_2$.

From Proposition 1, we derive that the firm will give up undertaking remanufacturing even if the production cost is relatively low (i.e., $c_r/\phi < c_n < (1 - e\beta_2)c_r/\phi(1 - e\beta_1)$). In particular, when $\beta_1 > \beta_2$, meaning DfE reduces the production cost of new products significantly more than it reduces the production cost of remanufactured products, the integrated firm becomes less willing to undertake remanufacturing. This result extends the frontier of the firm's remanufacturing compared to the case without DfE (as shown in Atasu et al. (2008) and Xiong et al. (2013), they found that the firm will not undertake remanufacturing when $c_r/\phi > c_n$). The intuition is that DfE reduces the production costs of new and remanufactured products and intensifies the competition between new and remanufactured products. In particular, when the DfE effort level is relatively high, the production cost of new products is significantly reduced since the marginal effect of DfE on new products' production cost is greater than that on remanufactured products' production cost. As a result, the integrated firm prefers to produce more new products. This will significantly intensify the competition between new and remanufactured products and results in remanufacturing hurting the integrated firm. Therefore, the integrated firm's willingness to engage in remanufacturing decreases with increments in the DfE effort level. The result is intuitive in the case where $\beta_1 < \beta_2$. In summary, with the presence of DfE, the boundary of the firm's remanufacturing willingness will change, which depends on the marginal effect of DfE on the production costs new and remanufactured products.

To better understand the results in Proposition 1, we provide Figure 2 in which $\beta_1 = 0.24$, $\beta_2 = 0.2$, $\phi = 0.42$ and $c_r = 0.3$ (or $c_n = 0.6$). As shown in Figure 2, when the DfE effort level increases, the area where the volume of remanufactured products equals zero expands, whereas the area where the volume of remanufactured products is positive shrinks. This result is consistent with the results presented above.

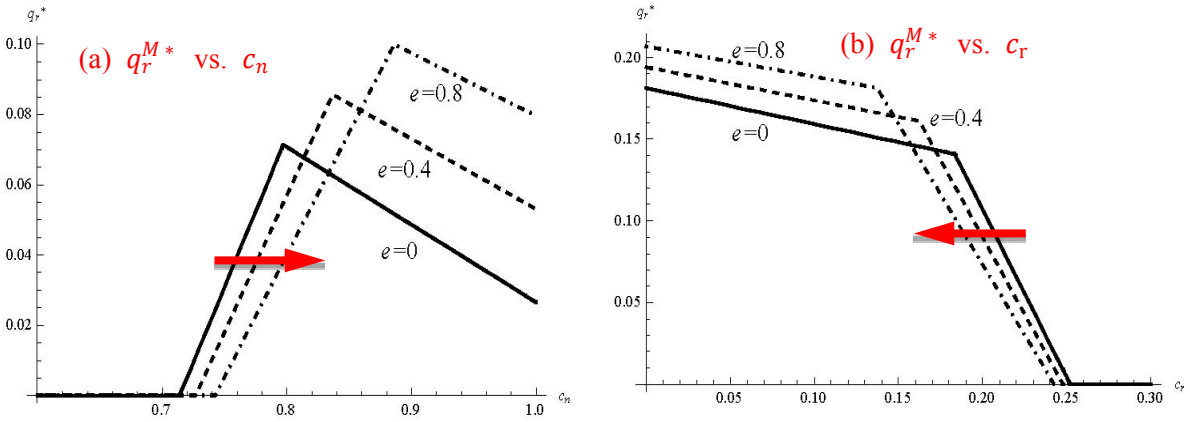


Figure 2. Optimal quantity of remanufactured products w.r.t. c_n and c_r. $(\beta_1 > \beta_2)$

4.2 Quantity decisions in a competitive case

In the competitive case, given the DfE effort level, OEM and IR simultaneously decide the quantities of new and remanufactured products, and a Nash game occurs between them. Similarly, the profit-maximization problems of OEM and IR can be expressed as

$$\pi_{OEM}^{C}(q_n|e) = q_n(p_n - \tilde{c}_n(e)) - F(e), \tag{5}$$

$$\pi_{IR}^{C}(q_r|e) = q_r(p_r - \tilde{c}_r(e)), \tag{6}$$

$$\text{s.t. } q_n \geq q_r \geq 0.$$

Analogously, given the DfE effort level, we derive optimal quantity decisions of the OEM and IR, which are given by Lemma 2.

Lemma 2. Given DfE effort level e, the optimal decisions of the OEM and IR are as follows:

(1) If $\frac{1-e\beta_1}{1-e\beta_2} \leq \frac{2c_r}{\phi c_n} - \frac{1}{c_n(1-e\beta_2)}$, then $q_n^{C*} = \frac{1-c_n(1-e\beta_1)}{2}$ and $q_r^{C*} = 0$.

(2) If $\frac{2c_r}{\phi c_n} - \frac{1}{c_n(1-e\beta_2)} \leq \frac{1-e\beta_1}{1-e\beta_2} \leq \frac{(2+\phi)c_r}{3\phi c_n} + \frac{1-\phi}{3c_n(1-e\beta_2)}$, then

$$q_n^{C*} = \frac{2-\phi-2c_n(1-e\beta_1)+c_r(1-e\beta_2)}{4-\phi} \text{ and } q_r^{C*} = \frac{\phi+\phi c_n(1-e\beta_1)-2c_r(1-e\beta_2)}{(4-\phi)\phi}.$$

(3) If $\frac{1-e\beta_1}{1-e\beta_2} \geq \frac{(2+\phi)c_r}{3\phi c_n} + \frac{1-\phi}{3c_n(1-e\beta_2)}$, then $q_r^{C*} = q_n^{C*} = \frac{1-c_n(1-e\beta_1)}{2+\phi}$.

Consequently, the corresponding profit of the OEM is summarized in Table 3.

Table 3. The OEM's profit in the competitive case

Conditions	$\pi_{OEM}^{C}(e)$
$\frac{1-e\beta_1}{1-e\beta_2} \le \frac{2c_r}{\phi c_n} - \frac{1}{c_n(1-e\beta_2)}$	$\frac{1-c_n(1-e\beta_1)(2-c_n(1-e\beta_1))}{4} - \frac{e^2k}{2}$
$\frac{2c_r}{\phi c_n} - \frac{1}{c_n(1-e\beta_2)} \le \frac{1-e\beta_1}{1-e\beta_2} \le \frac{(2+\phi)c_r}{3\phi c_n} + \frac{1-\phi}{3c_n(1-e\beta_2)}$	$\frac{2(2-\phi)^2+2c_r(1-e\beta_2)(2(2-\phi)+c_r(1-e\beta_2))}{+8c_n^2(1-e\beta_1)^2-8c_n(1-e\beta_1)(2-\phi+c_r(1-e\beta_2))}{2(4-\phi)^2} - \frac{e^2k}{2}$
$\frac{1-e\beta_1}{1-e\beta_2} \ge \frac{(2+\phi)c_r}{3\phi c_n} + \frac{1-\phi}{3c_n(1-e\beta_2)}$	$\frac{2-2c_n(1-e\beta_1)(2-c_n(1-e\beta_1))}{2(2+\phi)^2} - \frac{e^2k}{2}$

Lemma 2 illustrates that the IR is prevented from entering the market when the DfE effort level exceeds a threshold value, i.e., $\phi \le 2c_r(1-e\beta_2) - \phi c_n(1-e\beta_1)$. This condition implies that the OEM's strategy of DfE can be regarded as a lever to compete with the IR. However, if the OEM's DfE effort level is low, then the IR would opt to produce remanufactured products of which the number equals to that of new products. As implied in Lemma 2, with the presence of DfE, the IR's willingness to undertake remanufacturing increases when $(1+c_n)\beta_2/c_n > \beta_1 > \beta_2$ and decreases otherwise (please refer to Proposition 2 in the following). In particular, with the presence of DfE, the IR enters the market and competes with the OEM if $2c_r/\phi c_n < ((1-e\beta_1)c_n+1)/c_n(1-e\beta_2)$.

Based on Lemma 2, we obtain the following corollary.

Proposition 2. In comparing the scenario without DfE (i.e., $e=0$), with the presence of DfE, the IR becomes more willing to enter the market and cannibalize the OEM's market share if $(1+c_n)\beta_2/c_n > \beta_1 > \beta_2$, and becomes less willing to enter the market if $\beta_1 > (1+c_n)\beta_2/c_n$.

The result in the case where $\beta_1 < \beta_2$ is intuitive. However, when $(1+c_n)\beta_2/c_n > \beta_1 > \beta_2$, the counterintuitive result that the IR is more willing to enter the market and cannibalize the OEM's market share. Intuitively, the IR should escape the market since the DfE decreases the production cost of new products more than the cost of remanufactured products (i.e., $\beta_1 > \beta_2$), which makes new products more competitive. However, we find that the threshold strictly decreases in DfE effort level e when $(1+c_n)\beta_2/c_n > \beta_1 > \beta_2$. In other words, when a positive DfE effort is exerted by the OEM, the IR is more likely to offer remanufactured products. Although DfE reduces the production cost of new products more significantly, it reduces the remanufacturing cost as well. In this case, the IR may still acquire the cost advantage when the production cost of new products is high. Therefore, the IR will be more likely to enter the market and compete with OEM when the production cost of new

products is high.

Similarly, Figure 3 is provided to help understand the results in Proposition 2 where the notations are $\beta_1 = 0.24$, $\beta_2 = 0.2$, $\phi = 0.42$ and $c_r = 0.3$ (or $c_n = 0.6$). As shown in Figure 3, as the DfE effort level increases, the area where the volume of remanufactured products equals zero shrinks, whereas the area where the volume of remanufactured products is positive expands. This result is consistent with the results presented above.

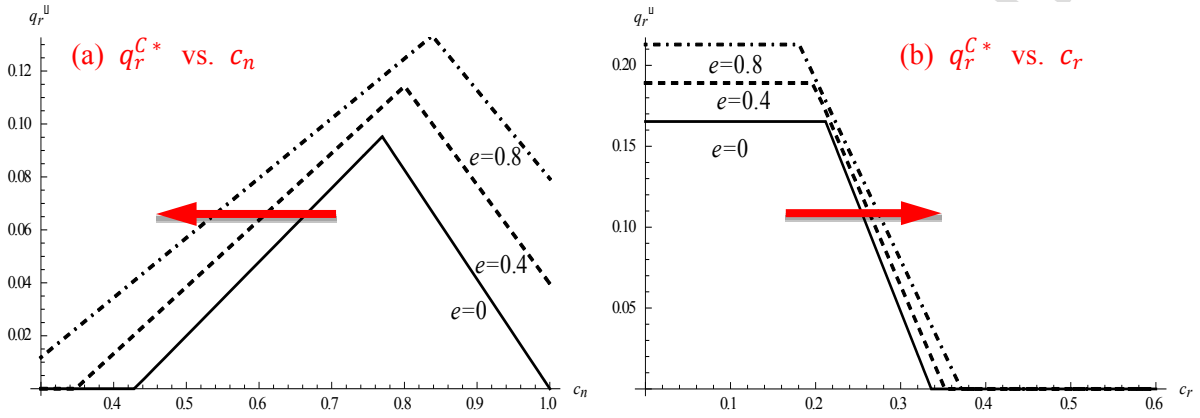


Figure 3. Optimal quantity of remanufactured product w.r.t. c_n and c_r.

Next, we perform a comparison to investigate the effects of DfE on firms' decisions. On the basis of Propositions 1 and 2, we obtain the following results.

Proposition 3. Given DfE effort level e, the optimal quantities in the two cases satisfy the following relationships.

(1) If $\frac{1-e\beta_1}{1-e\beta_2} \le \frac{2c_r}{\phi c_n} - \frac{1}{c_n(1-e\beta_2)}$, then $q_n^{M*} = q_n^{C*}$ and $q_r^{M*} = q_r^{C*}$.

(2) If $\frac{2c_r}{\phi c_n} - \frac{1}{c_n(1-e\beta_2)} \le \frac{1-e\beta_1}{1-e\beta_2} \le \frac{(2+\phi)c_r}{3\phi c_n} + \frac{1-\phi}{3c_n(1-e\beta_2)}$, then $q_n^{M*} > q_n^{C*}$ and $q_r^{M*} < q_r^{C*}$.

(3) If $\frac{(2+\phi)c_r}{3\phi c_n} + \frac{1-\phi}{3c_n(1-e\beta_2)} < \frac{1-e\beta_1}{1-e\beta_2} < \frac{1-\phi}{2c_n(1-e\beta_2)} + \frac{(1+\phi)c_r}{2\phi c_n}$, then $q_n^{M*} < q_n^{C*}$ and $q_r^{M*} < q_r^{C*}$.

(4) If $\frac{1-\phi}{2c_n(1-e\beta_2)} + \frac{(1+\phi)c_r}{2\phi c_n} < \frac{1-e\beta_1}{1-e\beta_2}$, then $q_r^{M*} = q_n^{M*} > q_r^{C*} = q_n^{C*}$ if $\frac{1-e\beta_1}{1-e\beta_2} > \frac{(3-\phi)}{5c_n(1-e\beta_2)} + \frac{(2+\phi)c_r}{5\phi c_n}$, and otherwise $q_r^{M*} = q_n^{M*} < q_r^{C*} = q_n^{C*}$ if $\frac{1-e\beta_1}{1-e\beta_2} < \frac{(3-\phi)}{5c_n(1-e\beta_2)} + \frac{(2+\phi)c_r}{5\phi c_n}$.

According to Proposition 3, when IR is prevented from entering the market, the quantities of new and remanufactured products are the same in the monopolistic and competitive cases. At this moment, the quantities of remanufactured products under both cases equal zero. Once the IR enters the market,

however, the quantity of remanufactured products in the competitive situation will be lower than that in the monopolistic situation if and only if the DfE effort level is low, i.e., $\frac{1-e\beta_1}{1-e\beta_2} > \frac{(3-\phi)}{5c_n(1-e\beta_2)} + \frac{(2+\phi)c_r}{5\phi c_n}$. Otherwise, the quantity of remanufactured products in the competitive situation will be higher than that in the monopolistic situation.

According to Proposition 3, with the presence of DfE, the number of new products in the monopolistic case may be larger (or smaller) than that in the competitive case, which is consistent with the existing works. However, under some conditions, the number of remanufactured products in the competitive case will be greater than that in the monopolistic case. This is because that under some conditions, the firm's willingness to undertake to remanufacture may reduce in the monopolistic situation but increase in the competitive situation with the presence of DfE (according to Propositions 1 and 2). Therefore, the number of remanufactured products in the competitive situation can be higher than that in the monopolistic case.

4.3 Impacts of DfE on the environment

In this section, we explore how DfE affects the environment. The total environmental impact in the monopolistic and competitive cases is provided in *Section II in the Appendix*. Several numerical examples can help examine the effect of DfE on the environment. The values of related parameters are as follows: $\rho_1 = 0.2$, $\rho_2 = 0.12$, $\beta_1 = 0.24$, $\beta_2 = 0.2$, $\delta_n = 0.3$, $\delta_r = 0.24$, $e = 0.8$and $\phi = 0.42$. The results are shown in Figure 4. Here, we just consider a case in which $\beta_1 > \beta_2$. In fact, the results still hold even if $\beta_1 < \beta_2$.

Figure 4 illustrates that DfE is not always beneficial to the environment, although it benefits the environment for a unit of product. Clearly, this total environmental impact depends on the production costs of new and remanufactured products, as well as the DfE effort level. When the production costs of new and remanufactured products are low, the total environmental impact can benefit from a higher level of DfE effort. The reason is that a higher level of DfE effort just slightly reduces the production costs when the production costs of new and remanufactured products are low, which consequently results in slightly increased sales of new and remanufactured products.

However, when the production costs of new and remanufactured products are high, a higher DfE effort level will be detrimental to the environment and aggravates the total environmental impact. This result is contrary to that of previous research on remanufacturing (Xiong et al. 2013; Zhang et al. 2018). The intuition is that a higher DfE can significantly reduce the production costs of new and remanufactured products, which leads to the significantly increased sales of new and remanufactured products. As a result, a dramatic increase in the production quantities results in the total environmental impact becoming worse off.

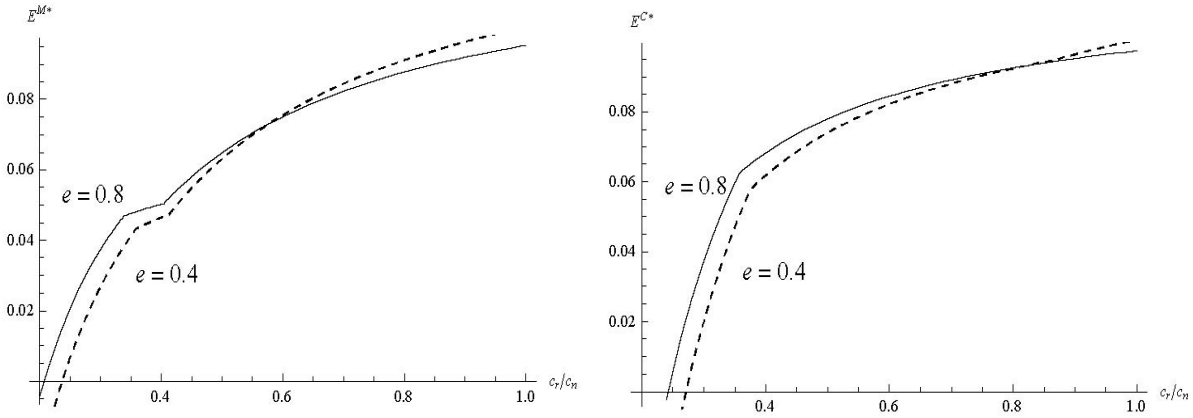


Figure 4. Total environmental impact E^{M*} and E^{C*} w.r.t. c_r/c_n. ($c_r = 0.3$)

5 Optimization of the DfE decision

In Section 4, given the DfE effort level, we explore firms' optimal quantity strategies in monopolistic and competitive cases. Several interesting outcomes are obtained. We also investigate OEM's optimal DfE strategy in this section. We let $\Delta = (\phi - c_r)/(1 - c_n)$, where Δ represents the relative profitability of remanufacturing compared with that of a new product.

5.1 DfE decision in the monopolistic case

When DfE is endogenous, the integrated firm in the monopolistic case determines the DfE effort level in the first stage and decides the quantities of new and remanufactured products in the second stage. According to Proposition 1, we obtain the optimal quantity decision in the second stage. Furthermore, we derive the optimal DfE effort level, which can be given by Proposition 4.

Proposition 4 In the monopolistic case, with the range of the scale parameter of investment function k, the optimal DfE effort level is as follows:

(i). If $\Delta \le \frac{2k\phi - \beta_1\beta_2 c_n c_r}{2k - c_n^2\beta_1^2}$, then $e^{M*} = \frac{(1-c_n)c_n\beta_1}{2k - c_n^2\beta_1^2}$, $q_r^{M*} = 0$ and $q_n^{M*} = \frac{1 - c_n(1 - e^{M*}\beta_1)}{2}$.

(ii). If $\frac{2k\phi - \beta_1\beta_2 c_n c_r}{2k - c_n^2\beta_1^2} < \Delta < \frac{4k\phi - c_r^2\beta_2^2}{2(1+\phi)k - c_n^2\beta_1^2 - \beta_1\beta_2 c_n c_r}$, then $e^{M*} = \frac{(1-\phi-c_n+c_r)\phi c_n\beta_1 + (\phi c_n - c_r)c_r\beta_2}{2(1-\phi)\phi(k - k_{\min})}$, $q_r^{M*} = \frac{\phi c_n(1 - e^{M*}\beta_1) - c_r(1 - e^{M*}\beta_2)}{2(1-\phi)\phi}$ and $q_n^{M*} = \frac{1 - \phi - c_n(1 - e^{M*}\beta_1) + c_r(1 - e^{M*}\beta_2)}{2(1-\phi)}$.

(iii). If $\Delta \ge \frac{4k\phi - c_r^2\beta_2^2}{2(1+\phi)k - c_n^2\beta_1^2 - \beta_1\beta_2 c_n c_r}$, then $e^{M*} = \frac{(1+\phi-c_n-c_r)(c_n\beta_1 + c_r\beta_2)}{2k(1+3\phi) - (c_n\beta_1 + c_r\beta_2)^2}$, and $q_r^{M*} = q_n^{M*} = \frac{1 + \phi - c_n(1 - e^{M*}\beta_1) - c_r(1 - e^{M*}\beta_2)}{2 + 6\phi}$.

When the profitability of remanufacturing is low, i.e., $\Delta \le \frac{2k\phi - \beta_1\beta_2 c_n c_r}{2k - c_n^2\beta_1^2}$, the integrated firm will exert a DfE effort so that the quantity of remanufactured products equals zero. In this case, the gains from selling remanufactured products will be less than the loss resulting from the cannibalization problem. Therefore, the integrated firm gives up remanufacturing. However, when remanufacturing is very profitable, i.e., $\Delta \ge \frac{4k\phi - c_r^2\beta_2^2}{2(1+\phi)k - c_n^2\beta_1^2 - \beta_1\beta_2 c_n c_r}$, the integrated firm prefers to exert a DfE effort so that the quantities of new and remanufactured products are the same. As remanufacturing is very profitable, the gain result from remanufacturing is higher than the corresponding loss, although the competition between new products and remanufactured products reaches its pinnacle in this case. When the profitability of remanufacturing is moderate, a DfE effort is exerted by the integrated firm so that the result "partial remanufacturing" is achieved.

5.2 DfE decision in the competitive case

In the competitive case, OEM decides the DfE effort level in the first stage, and the OEM and IR simultaneously decide the quantities of new and remanufactured products in the second stage. A Nash game occurs between the OEM and IR. In accordance with Proposition 2, we obtain the optimal quantity decision in the second stage. Furthermore, we derive the optimal DfE effort level, which can be given by Proposition 5.

Proposition 5 In the competitive case, with the range of the scale parameter of investment function k, the optimal DfE effort level is as follows:

(1) If $\Delta \le \frac{k\phi - \beta_1\beta_2 c_n c_r}{2k - c_n^2\beta_1^2}$, then $e^{C*} = \frac{(1-c_n)c_n\beta_1}{2k - c_n^2\beta_1^2}$, $q_r^{C*} = 0$ and $q_n^{C*} = \frac{1 - c_n(1 - e^{C*}\beta_1)}{2}$.

(2) If $\frac{k\phi-\beta_1\beta_2c_nc_r}{2k-c_n^2\beta_1^2}\le\Delta\le\frac{(4-\phi)\phi k-2(2c_n\beta_1-c_r\beta_2)c_r\beta_2}{2(4-\phi)k-4{\beta_1}^2{c_n}^2+2\beta_1\beta_2c_nc_r}$, then $e^{C*}=\frac{\phi+\phi c_n-2c_r}{\phi c_n\beta_1-2c_r\beta_2}$, $q_r^{C*}=0$ and $q_n^{C*}=\frac{2-\phi-2c_n(1-e^{C*}\beta_1)+c_r(1-e^{C*}\beta_2)}{4-\phi}$.

(3) If $\frac{(4-\phi)\phi k-2(2c_n\beta_1-c_r\beta_2)c_r\beta_2}{2(4-\phi)k-4{\beta_1}^2{c_n}^2+2\beta_1\beta_2c_nc_r}\le\Delta\le 2-\sqrt{\frac{k(4-\phi)^2-2(2c_n\beta_1-c_r\beta_2)^2}{k(2+\phi)^2-2c_n^2\beta_1^2}}$, then

$e^{C*}=\frac{2(2-\phi-2c_n+c_r)(2c_n\beta_1-c_r\beta_2)}{k(4-\phi)^2-2(2c_n\beta_1-c_r\beta_2)^2}$, $q_r^{C*}=\frac{\phi+\phi c_n(1-e^{C*}\beta_1)-2c_r(1-e^{C*}\beta_2)}{(4-\phi)\phi}$ and $q_n^{C*}=\frac{2-\phi-2c_n(1-e^{C*}\beta_1)+c_r(1-e^{C*}\beta_2)}{4-\phi}$.

(4) If $\Delta\ge 2-\sqrt{\frac{k(4-\phi)^2-2(2c_n\beta_1-c_r\beta_2)^2}{k(2+\phi)^2-2c_n^2\beta_1^2}}$, then $e^{C*}=\frac{2(1-c_n)c_n\beta_1}{k(2+\phi)^2-2c_n^2\beta_1^2}$, $q_r^{C*}=q_n^{C*}=\frac{1-c_n(1-e^{C*}\beta_1)}{2+\phi}$.

Proposition 5 indicates that the OEM cannot prevent the IR from entering the market in a competitive case if the profitability of remanufacturing is high, i.e., $\Delta\ge\frac{(4-\phi)\phi k-2(2c_n\beta_1-c_r\beta_2)c_r\beta_2}{2(4-\phi)k-4{\beta_1}^2{c_n}^2+2\beta_1\beta_2c_nc_r}$. In this case, although the OEM possesses an additional lever (i.e., DfE) in response to the IR entry threat, it fails to prevent the IR from entering the market. However, if the profitability of remanufacturing is low, i.e., $\Delta\le\frac{(4-\phi)\phi k-2(2c_n\beta_1-c_r\beta_2)c_r\beta_2}{2(4-\phi)k-4{\beta_1}^2{c_n}^2+2\beta_1\beta_2c_nc_r}$, the IR will never enter the market and cannibalize the OEM's market share.

An interesting result is that the IR will be prevented from entering the market when remanufacturing is relatively profitable, i.e., $\frac{k\phi-\beta_1\beta_2c_nc_r}{2k-c_n^2\beta_1^2}\le\Delta\le\frac{(4-\phi)\phi k-2(2c_n\beta_1-c_r\beta_2)c_r\beta_2}{2(4-\phi)k-4{\beta_1}^2{c_n}^2+2\beta_1\beta_2c_nc_r}$. In this case, the OEM can improve the competitive advantage of new products by exerting an appropriate DfE effort. Thus, DfE serves as a powerful lever to help the OEM compete with the potential IR.

To investigate how competition affects OEM's optimal DfE decisions, we provide comparisons through numerical examples because many situations with sophisticated derivation processes are involved. By setting $c_n=0.6$, $\beta_1=0.6$, $\beta_2=0.1$, $c_r=0.3$, and $\Delta e^*=e^{M*}-e^{C*}$, we can obtain Figure 5 by changing the values of k and ϕ. Clearly, a high value of ϕ implies that the profitability of remanufacturing is high. We obtain the following conclusions by examining Figure 5.

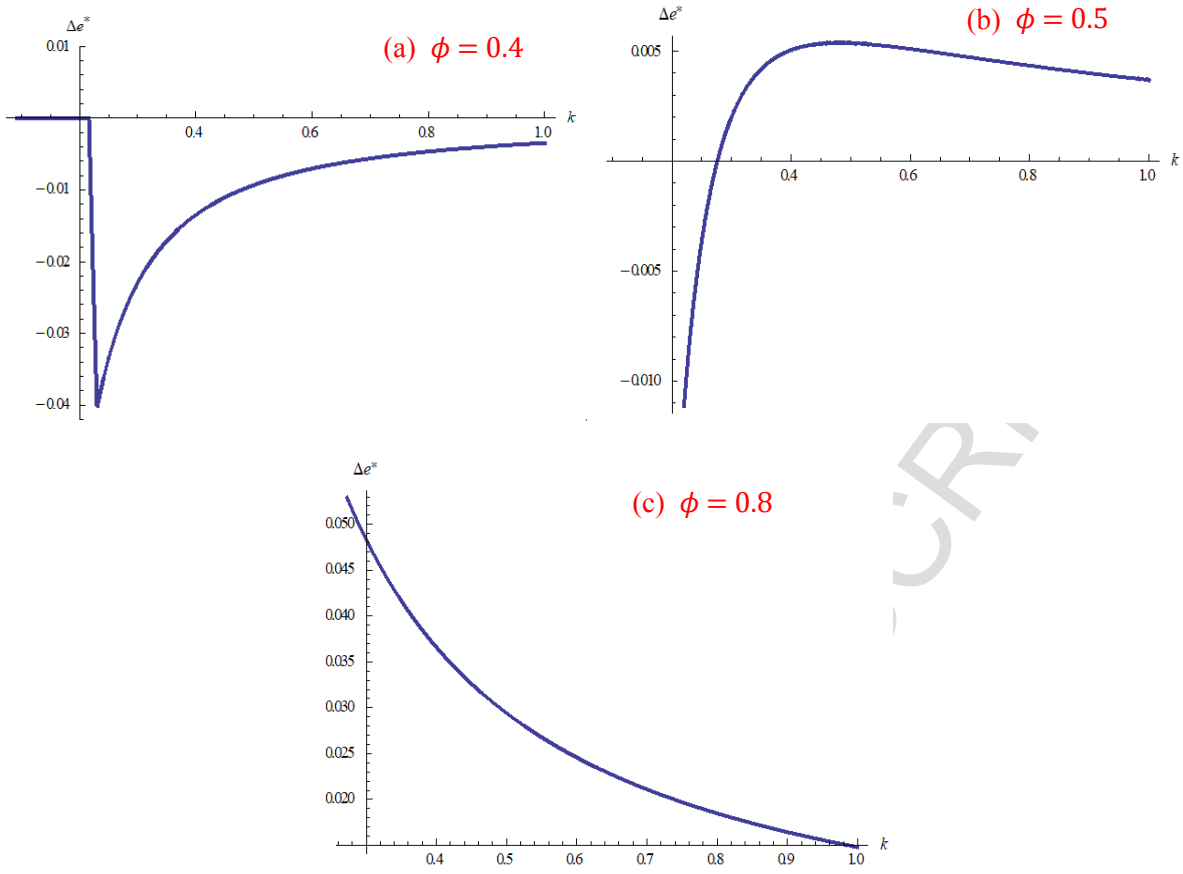


Figure 5 Optimal DfE effort level w.r.t. k and ϕ

Observation 1 Thresholds $\hat{\phi}$ and $\hat{k}$ exist, in which the optimal DfE effort level in the monopolistic case is lower than that in the competitive case, that is, if $\hat{\phi} > \phi$ and $k < \hat{k}$, and $0.4 < \hat{\phi} < 0.5$, then $e^{M*} \leq e^{C*}$.

Observation 1 shows that the OEM's optimal DfE effort level in the competitive situation is higher than that in the monopolistic situation under certain conditions. The OEM has many incentives to devote high DfE effort when the scale parameter of the investment function is relatively small. At the same time, the competition between new and remanufactured products is low when the profitability of remanufacturing is low (i.e., $\hat{\phi} > \phi$). In the competitive environment, the OEM reduces the cost disadvantage of new products because a high DfE effort level can significantly reduce the production cost. Therefore, the OEM has a great motivation to improve the level of its DfE effort in the competitive circumstance.

Observation 2 Thresholds $\hat{\bar{\phi}}$ and $\hat{\bar{k}}$ exist, in which the optimal DfE effort level in the monopolistic case is higher than that in the competitive case, that is, if $\hat{\bar{\phi}} < \phi$ and $k > \hat{\bar{k}}$, and $0.4 <$

$\hat{\bar{\phi}} < 0.5$, then $e^{M*} > e^{C*}$.

Observation 2 shows that the OEM's optimal DfE effort level in the monopolistic situation is higher than that in the competitive situation under certain conditions. The OEM has fewer incentives to devote a high DfE effort level when the scale parameter of the investment function is relatively large. In addition, the competition between new and remanufactured products is high when the profitability of remanufacturing is high (i.e., $\hat{\bar{\phi}} < \phi$). In a competitive environment, the OEM prefers to exert a low DfE effort level to avoid fierce competition and the cannibalization problem. Therefore, the OEM has less motivation to improve its DfE effort level in the competitive circumstance.

6 Conclusions

Increments in consumer awareness of environmental protection encourage firms to engage in reverse logistics management and recycling and restoring old products through remanufacturing. Furthermore, firms have begun to incorporate product remanufacturing into product design because related environmental laws have become more stringent. Through DfE, firms can develop recycling and remanufacturing technologies to make products environmentally-friendly and easy to remanufacture. Given this background, a theoretical model is constructed in this study. The model effectively captures the essential feature of DfE, that is, DfE can simultaneously reduce production and remanufacturing costs. We discuss two cases to study the effects of DfE on firms' remanufacturing willingness and environmental impact and how competition influences firms' DfE strategies. By analyzing and comparing the optimal decisions, we obtain the following conclusions.

(1) In a monopolistic situation, DfE reduces a firm's willingness to engage in remanufacturing. In a competitive situation, however, DfE encourages the IR to enter the market and erode the OEM's market share.

(2) With regard to the impact of DfE on environmental impact, DfE may be detrimental to the environment although it helps reduce the environmental impacts of a unit of product. The specific result relies on the baseline production costs of new and remanufactured products.

(3) When the profitability of remanufacturing is low, and the scale parameter of the investment function is small, a firm's DfE effort in a competitive case is higher than that in a monopolistic case.

When the profitability of remanufacturing is high, and the scale parameter of the investment function is large, the level of DfE effort exerted by the firm in a monopolistic environment is higher than that in a competitive environment.

We consider the influences of DfE on firms' remanufacturing decisions and environment and investigate how competition affects firms' DfE strategies. Future research can start from the perspective of the supply chain and analyze firms' optimal strategies of DfE in a vertical competition environment.

Acknowledgements

Appendix

Section I

Proof of Lemma 1

Proof: According to Equations (1), (2) and (4), the integrated firm's optimization problem can be given by

$$\pi_I^M(q_n,q_r|e) = q_n(1-q_n-\phi q_r-(1-\beta_1 e)c_n)+q_r(\phi(1-q_n-q_r)-(1-\beta_2 e)c_r)-\frac{ke^2}{2},$$
$$\text{s.t. } q_n \geq q_r \geq 0.$$

Given the DfE level e, it is simple to prove that $\pi_I^M(q_n,q_r|e)$ is jointly concave in q_n and q_r. Furthermore, the Lagrange formulation can be expressed by

$$L(q_n,q_r,u,v) = q_n(1-q_n-\phi q_r-(1-\beta_1 e)c_n)+q_r(\phi(1-q_n-q_r)-(1-\beta_2 e)c_r)-\frac{ke^2}{2}+u(q_n-q_r)+vq_r.$$

According to the Kuhn-Tucker Theorem, we have

$$\frac{\partial L}{\partial q_n} = 1+u-2q_n-2\phi q_r+c_n(-1+e\beta_1)=0,$$
$$\frac{\partial L}{\partial q_r} = -u+v+\phi-2\phi(q_n+q_r)+c_r(-1+e\beta_2),$$
$$u(q_n-q_r)=0,$$
$$vq_r=0,$$
$$u,v\geq 0.$$

Obviously, the Lagrange multiplier u,v cannot be simultaneously positive, therefore there are only three cases. In particular, we have

(i) If $u=0$ and $v\neq 0$, we have $q_n^{M*}=\frac{1-c_n(1-e\beta_1)}{2}$, $q_r^{M*}=0$ and $v=c_r(1-e\beta_2)-\phi c_n(1-e\beta_1)$. From $v>0$, we can obtain $\frac{1-e\beta_1}{1-e\beta_2}\leq\frac{c_r}{\phi c_n}$.

(ii) If $u=0$ and $v=0$, we have $q_r^{M*}=\frac{\phi c_n(1-e\beta_1)-c_r(1-e\beta_2)}{2(1-\phi)\phi}$ and $q_n^{M*}=\frac{1-\phi-c_n(1-e\beta_1)+c_r(1-e\beta_2)}{2(1-\phi)}$. Since $0<q_r<q_n$, we have $\frac{c_r}{\phi c_n}<\frac{1-e\beta_1}{1-e\beta_2}<\frac{1-\phi}{2c_n(1-e\beta_2)}+\frac{(1+\phi)c_r}{2\phi c_n}$.

(iii) If $u\neq 0$ and $v=0$, we have $q_r^{M*}=q_n^{M*}=\frac{1+\phi-c_n(1-e\beta_1)-c_r(1-e\beta_2)}{2+6\phi}$ and $u=\frac{2\phi c_n(1-e\beta_1)-(1-e\beta_2)(1+\phi)c_r-(1-\phi)\phi}{1+3\phi}$. From $u>0$, we can obtain $\frac{1-e\beta_1}{1-e\beta_2}\geq\frac{1-\phi}{2c_n(1-e\beta_2)}+\frac{(1+\phi)c_r}{2\phi c_n}$.

Proof of Proposition 1

Proof: To prove proposition 1, we just need to explore that the monotonicity of the threshold $\frac{1-e\beta_1}{1-e\beta_2}$ with respect to e. By taking the first order derivative of $\frac{1-e\beta_1}{1-e\beta_2}$ with respect to e, we have $\frac{\partial}{\partial e}\left(\frac{1-e\beta_1}{1-e\beta_2}\right)=\frac{\beta_2-\beta_1}{(1-e\beta_2)^2}<0$ if $\beta_2<\beta_1$ and $\frac{\partial}{\partial e}\left(\frac{1-e\beta_1}{1-e\beta_2}\right)=\frac{\beta_2-\beta_1}{(1-e\beta_2)^2}>0$ if $\beta_2>\beta_1$.

Proof of Lemma 2

Proof: According to Equations (1), (2), (5) and (6), we have

$$\pi_{OEM}^{C}(q_n|e)=q_n(1-q_n-\phi q_r-(1-\beta_1 e)c_n)-\frac{ke^2}{2},$$
$$\pi_{IR}^{C}(q_r|e)=q_r(\phi(1-q_n-q_r)-(1-\beta_2 e)c_r),$$
$$\text{s.t. } q_n\geq q_r\geq 0.$$

First of all, we can prove that $\pi_{OEM}^{C}(q_n|e)$ is concave in q_n, and $\pi_{IR}^{C}(q_r|e)$ is concave in q_r. Therefore, we can acquire the OEM's and IR's optimal quantity decisions.

$$q_n(q_r)=\frac{1-\phi q_r-c_n(1-e\beta_1)}{2} \text{ and } q_r(q_n)=\frac{\phi-\phi q_n-c_r(1-e\beta_2)}{2\phi}.$$

According to $q_n\geq q_r\geq 0$, then we can obtain the results below.

(1) If $q_r=\frac{\phi-\phi q_n-c_r(1-e\beta_2)}{2\phi}\leq 0$, i.e., $q_r^*=0$, then we get $q_n^*=\frac{1-c_n(1-e\beta_1)}{2}$. From $q_r=q_r(q_n^*)\leq 0$, we can derive $\frac{1-e\beta_1}{1-e\beta_2}\leq\frac{2c_r}{\phi c_n}-\frac{1}{c_n(1-e\beta_2)}$.

(2) If $0<q_r<q_n$, i.e., $q_r^*=q_r(q_n)$, by solving $q_n=\frac{1-\phi q_r-c_n(1-e\beta_1)}{2}$ and $q_r(q_n)=\frac{\phi-\phi q_n-c_r(1-e\beta_2)}{2\phi}$, we obtain

$$q_n^*=\frac{2-\phi-2c_n(1-e\beta_1)+c_r(1-e\beta_2)}{4-\phi} \text{ and } q_r^*=\frac{\phi+\phi c_n(1-e\beta_1)-2c_r(1-e\beta_2)}{(4-\phi)\phi}$$

Furthermore, since $0 < q_r < q_n$, we have $\frac{2c_r}{\phi c_n} - \frac{1}{c_n(1-e\beta_2)} \le \frac{1-e\beta_1}{1-e\beta_2} \le \frac{(2+\phi)c_r}{3\phi c_n} + \frac{1-\phi}{3c_n(1-e\beta_2)}$.

(3) If $q_n \le q_r = \frac{\phi - \phi q_n - c_r(1-e\beta_2)}{2\phi}$, i.e., $q_r^* = q_n$, then we can obtain $q_r^* = q_n^* = \frac{1-c_n(1-e\beta_1)}{2+\phi}$.

From $0 < q_n^* \le q_r^*$, we can get $\frac{1-e\beta_1}{1-e\beta_2} \ge \frac{(2+\phi)c_r}{3\phi c_n} + \frac{1-\phi}{3c_n(1-e\beta_2)}$.

Proof of Proposition 2

Proof: Similar to the proof of Proposition 1, we need to explore that the monotonicity of the threshold $\frac{1-e\beta_1}{1-e\beta_2} + \frac{1}{c_n(1-e\beta_2)}$ with respect to e. By taking the first derivative of $\frac{1-e\beta_1}{1-e\beta_2} + \frac{1}{c_n(1-e\beta_2)}$ respect to e, we have $\frac{\partial}{\partial e}\left(\frac{1-e\beta_1}{1-e\beta_2} + \frac{1}{c_n(1-e\beta_2)}\right) = \frac{\beta_2-\beta_1}{(1-e\beta_2)^2} + \frac{\beta_2}{c_n(1-e\beta_2)^2}$. Therefore, we can get $\frac{\partial}{\partial e}\left(\frac{1-e\beta_1}{1-e\beta_2} + \frac{1}{c_n(1-e\beta_2)}\right) = \frac{\beta_2-\beta_1}{(1-e\beta_2)^2} + \frac{\beta_2}{c_n(1-e\beta_2)^2} > 0$ if $\beta_2 < \beta_1 < (1+c_n)\beta_2/c_n$, and $\frac{\partial}{\partial e}\left(\frac{1-e\beta_1}{1-e\beta_2} + \frac{1}{c_n(1-e\beta_2)}\right) = \frac{\beta_2-\beta_1}{(1-e\beta_2)^2} + \frac{\beta_2}{c_n(1-e\beta_2)^2} < 0$ otherwise.

Proof of Proposition 3

Proof: According to Propositions 1 and 2, we have

$$\frac{2c_r}{\phi c_n} - \frac{1}{c_n(1-e\beta_2)} < \frac{c_r}{\phi c_n} \text{ and } \frac{(2+\phi)c_r}{3\phi c_n} + \frac{1-\phi}{3c_n(1-e\beta_2)} < \frac{1-\phi}{2c_n(1-e\beta_2)} + \frac{(1+\phi)c_r}{2\phi c_n}.$$

Therefore, we have

(i). If $\frac{1-e\beta_1}{1-e\beta_2} \le \frac{2c_r}{\phi c_n} - \frac{1}{c_n(1-e\beta_2)}$, then we have $q_r^{M*} = q_r^{C*}$ and $q_n^{M*} = q_n^{C*}$.

(ii). If $\frac{2c_r}{\phi c_n} - \frac{1}{c_n(1-e\beta_2)} \le \frac{1-e\beta_1}{1-e\beta_2} \le \frac{c_r}{\phi c_n}$, we can derive

$$q_r^{M*} - q_r^{C*} = -q_r^{C*} < 0 \text{ and } q_n^{M*} - q_n^{C*} = \frac{\phi + \phi c_n(1-e\beta_1) - 2c_r(1-e\beta_2)}{2(4-\phi)} \ge 0.$$

(iii). If $\frac{c_r}{\phi c_n} < \frac{1-e\beta_1}{1-e\beta_2} < \frac{(2+\phi)c_r}{3\phi c_n} + \frac{1-\phi}{3c_n(1-e\beta_2)}$, we get

$$q_r^{M*} - q_r^{C*} = \frac{(2+\phi)c_n(1-e\beta_1) - 2(1-\phi) - 3c_r(1-e\beta_2)}{2(4-\phi)(1-\phi)} \le \frac{c_r(1-e\beta_2) - \phi}{6\phi} < 0 \text{ and } q_n^{M*} - q_n^{C*} = \frac{(1-\phi)\phi - 3\phi c_n(1-e\beta_1) + (2+\phi)c_r(1-e\beta_2)}{2(4-\phi)(1-\phi)} \ge 0.$$

(iv). If $\frac{(2+\phi)c_r}{3\phi c_n} + \frac{1-\phi}{3c_n(1-e\beta_2)} \le \frac{1-e\beta_1}{1-e\beta_2} < \frac{1-\phi}{2c_n(1-e\beta_2)} + \frac{(1+\phi)c_r}{2\phi c_n}$, we have

$$q_r^{M*} - q_r^{C*} = \frac{(4-\phi)\phi c_n(1-e\beta_1) - 2(1-\phi)\phi - (2+\phi)c_r(1-e\beta_2)}{2(1-\phi)(2+\phi)\phi} \le \frac{c_r(1-e\beta_2) - \phi}{4(2+\phi)c_n} < 0 \text{ and } q_n^{M*} - q_n^{C*} = \frac{(1-\phi)\phi - 3\phi c_n(1-e\beta_1) + (2+\phi)c_r(1-e\beta_2)}{2(2-\phi-\phi^2)} \le 0.$$

(v). If $\frac{1-\phi}{2c_n(1-e\beta_2)}+\frac{(1+\phi)c_r}{2\phi c_n}\leq\frac{1-e\beta_1}{1-e\beta_2}$, we have

$$q_r^{M*}-q_r^{C*}=q_n^{M*}-q_n^{C*}=\frac{5\phi c_n(1-e\beta_1)-(3-\phi)\phi-(2+\phi)c_r(1-e\beta_2)}{2(2+\phi)(1+3\phi)}.$$

Since $q_r^{M*}-q_r^{C*}$ is monotonic in e, we can derive

$$q_r^{M*}-q_r^{C*}=q_n^{M*}-q_n^{C*}=\frac{5\phi c_n(1-e\beta_1)-(3-\phi)\phi-(2+\phi)c_r(1-e\beta_2)}{2(2+\phi)(1+3\phi)}<0 \text{ if } \frac{1-e\beta_1}{1-e\beta_2}<\frac{(3-\phi)}{5c_n(1-e\beta_2)}+\frac{(2+\phi)c_r}{5\phi c_n}.$$

Proof of Proposition 4

Proof: Based on Proposition 1, we can obtain the following outcomes.

(i) If $\frac{1-e\beta_1}{1-e\beta_2}\leq\frac{c_r}{\phi c_n}$, we have $\pi_I^M(e)=\frac{1-2e^2k-c_n(1-e\beta_1)(2-c_n(1-e\beta_1))}{4}$.

Furthermore, by taking the second order derivative, we have

$$\frac{\partial^2\pi_I^M}{\partial e^2}=-k+\frac{1}{2}c_n^2\beta_1^2<0\Leftrightarrow k>\frac{c_n^2\beta_1^2}{2}.$$

Therefore, $\pi_I^M(e)$ has the unique maximal solution. According to the first order derivative condition, we have

$$e^{M*}=\frac{(1-c_n)c_n\beta_1}{2k-c_n^2\beta_1^2} \text{ and } \frac{(1-c_n)c_n\beta_1}{2k-c_n^2\beta_1^2}\geq\frac{\phi c_n-c_r}{\beta_1\phi c_n-\beta_2 c_r}\Leftrightarrow\frac{\phi-c_r}{1-c_n}\leq\frac{2k\phi-\beta_1\beta_2c_nc_r}{2k-c_n^2\beta_1^2}.$$

(ii) If $\frac{c_r}{\phi c_n}<\frac{1-e\beta_1}{1-e\beta_2}<\frac{1-\phi}{2c_n(1-e\beta_2)}+\frac{(1+\phi)c_r}{2\phi c_n}$, we have

$$\pi_I^M(e)=\frac{(1-2e^2k)(1-\phi)\phi+\phi c_n^2(1-e\beta_1)^2+c_r^2(1-e\beta_2)^2-2\phi c_n(1-e\beta_1)(1-\phi+c_r(1-e\beta_2))}{4(1-\phi)\phi}.$$

Similarly, we can derive $\frac{\partial^2\pi_I^M}{\partial e^2}<0$ if $k>\frac{\phi c_n^2\beta_1^2-2\phi c_nc_r\beta_1\beta_2+c_r^2\beta_2^2}{2(1-\phi)\phi}=k_{\min}$. Furthermore, we have $e^{M*}=\frac{(1-\phi-c_n+c_r)\phi c_n\beta_1+(\phi c_n-c_r)c_r\beta_2}{2k(1-\phi)\phi-(\phi c_n^2\beta_1^2-2\phi c_nc_r\beta_1\beta_2+c_r^2\beta_2^2)}$, and $\frac{c_r}{\phi c_n}<\frac{1-e\beta_1}{1-e\beta_2}<\frac{1-\phi}{2c_n(1-e\beta_2)}+\frac{(1+\phi)c_r}{2\phi c_n}\Leftrightarrow\frac{2k\phi-\beta_1\beta_2c_nc_r}{2k-c_n^2\beta_1^2}\leq\frac{\phi-c_r}{1-c_n}\leq\frac{4k\phi-c_r^2\beta_2^2}{2(1+\phi)k-c_n^2\beta_1^2-\beta_1\beta_2c_nc_r}$.

(iii) If $\frac{1-e\beta_1}{1-e\beta_2}\geq\frac{1-\phi}{2c_n(1-e\beta_2)}+\frac{(1+\phi)c_r}{2\phi c_n}$, we have $\pi_I^M(e)=\frac{(1+\phi-c_n(1-e\beta_1)-c_r(1-e\beta_2))^2}{4(1+3\phi)}-\frac{e^2k}{2}$.

Analogously, we have

$$\frac{\partial^2\pi_I^M}{\partial e^2}=\frac{-2k(1+3\phi)+(c_n\beta_1+c_r\beta_2)^2}{2(1+3\phi)}<0\Leftrightarrow k>\frac{(c_n\beta_1+c_r\beta_2)^2}{2(1+3\phi)}.$$

$$e^{M*}=\frac{(1+\phi-c_n-c_r)(c_n\beta_1+c_r\beta_2)}{2k(1+3\phi)-(c_n\beta_1+c_r\beta_2)^2} \text{ and } \frac{1-e\beta_1}{1-e\beta_2}\geq\frac{1-\phi}{2c_n(1-e\beta_2)}+\frac{(1+\phi)c_r}{2\phi c_n}\Leftrightarrow\frac{\phi-c_r}{1-c_n}\geq\frac{4k\phi-c_r^2\beta_2^2}{2(1+\phi)k-c_n^2\beta_1^2-\beta_1\beta_2c_nc_r}.$$

Proof of Proposition 5

Proof: Based on Proposition 2, we have

(1) When $\frac{1+c_n(1-e\beta_1)}{2} \le \frac{c_r(1-e\beta_2)}{\phi}$, we have $\pi_{OEM}^C(e) = \frac{1-2e^2k-c_n(1-e\beta_1)(2-c_n(1-e\beta_1))}{4}$.

Furthermore, by taking the second order derivative, we have

$$\frac{\partial^2 \pi_{OEM}^C}{\partial e^2} = -k + \frac{c_n^2\beta_1^2}{2} < 0 \Leftrightarrow k > \frac{c_n^2\beta_1^2}{2}.$$

Therefore, $\pi_{OEM}^C(e)$ has the unique maximal solution. According to the first order derivative condition, we have

$$e^{C*} = \frac{(1-c_n)c_n\beta_1}{2k-c_n^2\beta_1^2} \text{ and } \frac{1+c_n(1-e^*\beta_1)}{2} \le \frac{c_r(1-e^*\beta_2)}{\phi} \Leftrightarrow \frac{\phi-c_r}{1-c_n} \le \frac{k\phi-\beta_1\beta_2c_nc_r}{2k-c_n^2\beta_1^2}.$$

(2) When $\frac{1+c_n(1-e\beta_1)}{2} > \frac{c_r(1-e\beta_2)}{\phi} > \frac{3c_n(1-e\beta_1)-(1-\phi)}{2+\phi}$, we have

$$\pi_{OEM}^C = \frac{2(2-\phi)^2+8c_n^2(1-e\beta_1)^2+2c_r(1-e\beta_2)(2(2-\phi)+c_r(1-e\beta_2))-8c_n(1-e\beta_1)(2-\phi+c_r(1-e\beta_2))-e^2k(4-\phi)^2}{2(4-\phi)^2}.$$

Similarly, we have

$$\frac{\partial^2 \pi_{OEM}^C}{\partial e^2} = \frac{-k(4-\phi)^2+2(2c_n\beta_1-c_r\beta_2)^2}{(4-\phi)^2} < 0 \Leftrightarrow k > \frac{2(2c_n\beta_1-c_r\beta_2)^2}{(4-\phi)^2}.$$

Furthermore, we have

$$e^{C*} = \frac{2(2-\phi-2c_n+c_r)(2c_n\beta_1-c_r\beta_2)}{k(4-\phi)^2-2(2c_n\beta_1-c_r\beta_2)^2} \text{ and } \frac{1+c_n(1-e^*\beta_1)}{2} > \frac{c_r(1-e^*\beta_2)}{\phi} > \frac{3c_n(1-e^*\beta_1)-(1-\phi)}{2+\phi} \Leftrightarrow$$

$$\frac{(4-\phi)\phi k-2(2c_n\beta_1-c_r\beta_2)c_r\beta_2}{2(4-\phi)k-4{\beta_1}^2{c_n}^2+2\beta_1\beta_2c_nc_r} \le \frac{\phi-c_r}{1-c_n} \le \frac{3(4-\phi)\phi k-2(2c_n\beta_1-c_r\beta_2)c_r\beta_2}{(2+\phi)(4-\phi)k-4{\beta_1}^2{c_n}^2+2\beta_1\beta_2c_nc_r}.$$

(3) When $\frac{3c_n(1-e\beta_1)-(1-\phi)}{2+\phi} \ge \frac{c_r(1-e\beta_2)}{\phi}$, we have $\pi_{OEM}^C = \frac{2-e^2k(2+\phi)^2-2c_n(1-e\beta_1)(2-c_n(1-e\beta_1))}{2(2+\phi)^2}$.

Analogously, we have

$$\frac{\partial^2 \pi_{OEM}^C}{\partial e^2} = -k + \frac{2c_n^2\beta_1^2}{(2+\phi)^2} < 0 \Leftrightarrow k > \frac{2c_n^2\beta_1^2}{(2+\phi)^2},$$

$$e^{C*} = \frac{2(1-c_n)c_n\beta_1}{k(2+\phi)^2-2c_n^2\beta_1^2} \text{ and } \frac{c_r(1-e\beta_2)}{\phi} \ge \frac{-1+\phi+3c_n(1-e\beta_1)}{2+\phi} \Leftrightarrow \frac{\phi-c_r}{1-c_n} \ge \frac{3(2+\phi)\phi k-2\beta_1\beta_2c_nc_r}{k(2+\phi)^2-2c_n^2\beta_1^2}.$$

In addition, we can prove that

$$\frac{k\phi-\beta_1\beta_2c_nc_r}{2k-c_n^2\beta_1^2} - \frac{(4-\phi)\phi k-2(2c_n\beta_1-c_r\beta_2)c_r\beta_2}{2(4-\phi)k-4{\beta_1}^2{c_n}^2+2\beta_1\beta_2c_nc_r} = \frac{-k(\phi c_n\beta_1-2c_r\beta_2)^2}{2(2k-c_n^2\beta_1^2)(k(4-\phi)-c_n\beta_1(2c_n\beta_1-c_r\beta_2))} < 0 \text{ and}$$

$$\frac{3(4-\phi)\phi k-2(2c_n\beta_1-c_r\beta_2)c_r\beta_2}{(2+\phi)(4-\phi)k-4{\beta_1}^2{c_n}^2+2\beta_1\beta_2c_nc_r} - \frac{3(2+\phi)\phi k-2\beta_1\beta_2c_nc_r}{(2+\phi)^2k-2{\beta_1}^2{c_n}^2} =$$

$$\frac{2k(3\phi c_n\beta_1-(2+\phi)c_r\beta_2)^2}{(k(2+\phi)^2-2c_n^2\beta_1^2)(k(4-\phi)(2+\phi)-2c_n\beta_1(2c_n\beta_1-c_r\beta_2))} > 0.$$

Therefore, the optimal solution will be the boundary point when $\frac{k\phi-\beta_1\beta_2c_nc_r}{2k-c_n^2\beta_1^2} \le \frac{\phi-c_r}{1-c_n} \le \frac{(4-\phi)\phi k-2(2c_n\beta_1-c_r\beta_2)c_r\beta_2}{2(4-\phi)k-4{\beta_1}^2{c_n}^2+2\beta_1\beta_2c_nc_r}$. Thus, we have

(1) If $\frac{\phi-c_r}{1-c_n} \le \frac{k\phi-\beta_1\beta_2c_nc_r}{2k-c_n^2\beta_1^2}$, then $e^{C*} = \frac{(1-c_n)c_n\beta_1}{2k-c_n^2\beta_1^2}$;

(2) If $\frac{k\phi-\beta_1\beta_2c_nc_r}{2k-c_n^2\beta_1^2} \le \frac{\phi-c_r}{1-c_n} \le \frac{(4-\phi)\phi k-2(2c_n\beta_1-c_r\beta_2)c_r\beta_2}{2(4-\phi)k-4{\beta_1}^2{c_n}^2+2\beta_1\beta_2c_nc_r}$, then $e^{C*} = \frac{\phi+\phi c_n-2c_r}{\phi c_n\beta_1-2c_r\beta_2}$;

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(3) If $\frac{(4-\phi)\phi k-2(2c_n\beta_1-c_r\beta_2)c_r\beta_2}{2(4-\phi)k-4{\beta_1}^2{c_n}^2+2\beta_1\beta_2c_nc_r}\leq\frac{\phi-c_r}{1-c_n}\leq 2-\sqrt{\frac{k(4-\phi)^2-2(2c_n\beta_1-c_r\beta_2)^2}{k(2+\phi)^2-2c_n^2\beta_1^2}}$, then

$$e^{C*}=\frac{2(2-\phi-2c_n+c_r)(2c_n\beta_1-c_r\beta_2)}{k(4-\phi)^2-2(2c_n\beta_1-c_r\beta_2)^2};$$

(4) If $\frac{\phi-c_r}{1-c_n}\geq 2-\sqrt{\frac{k(4-\phi)^2-2(2c_n\beta_1-c_r\beta_2)^2}{k(2+\phi)^2-2c_n^2\beta_1^2}}$, then $e^{C*}=\frac{2(1-c_n)c_n\beta_1}{k(2+\phi)^2-2c_n^2\beta_1^2}$,

where $2-\sqrt{\frac{k(4-\phi)^2-2(2c_n\beta_1-c_r\beta_2)^2}{k(2+\phi)^2-2c_n^2\beta_1^2}}$ is the positive root of $\pi_{OEM}(e=\frac{2(2-\phi-2c_n+c_r)(2c_n\beta_1-c_r\beta_2)}{k(4-\phi)^2-2(2c_n\beta_1-c_r\beta_2)^2})=\pi_{OEM}(e=\frac{2(1-c_n)c_n\beta_1}{k(2+\phi)^2-2c_n^2\beta_1^2})$.

Section II

In this section, we explore how DfE affects the environmental impact. Based on Equation (3) and Proposition 1, we can obtain the environmental impact in the monopolistic case, which can be given by Lemma 1.

Lemma 1. Given the DfE effort level e, the environmental impact in the monopolistic case is

(1) If $\frac{1-e\beta_1}{1-e\beta_2}\leq\frac{c_r}{\phi c_n}$, then $E^{M*}=\frac{(1-\rho_1e)\delta_n(1-c_n(1-e\beta_1))}{2}$;

(2) If $\frac{c_r}{\phi c_n}<\frac{1-e\beta_1}{1-e\beta_2}<\frac{1-\phi}{2c_n(1-e\beta_2)}+\frac{(1+\phi)c_r}{2\phi c_n}$, then

$$E^{M*}=\frac{(1-\rho_1e)\delta_n(\phi c_n(1-e\beta_1)-c_r(1-e\beta_2))}{2(1-\phi)\phi}+\frac{(1-\rho_2e)\delta_r(1-\phi-c_n(1-e\beta_1)+c_r(1-e\beta_2))}{2(1-\phi)}.$$

(3) If $\frac{1-e\beta_1}{1-e\beta_2}\geq\frac{1-\phi}{2c_n(1-e\beta_2)}+\frac{(1+\phi)c_r}{2\phi c_n}$, then $E^{M*}=\frac{((1-\rho_1e)\delta_n+(1-\rho_2e)\delta_r)(1+\phi-c_n(1-e\beta_1)-c_r(1-e\beta_2))}{2+6\phi}$.

Similarly, according to Equation (3) and Proposition 2, we can obtain the environmental impact in the competitive case, which can be given by Lemma 2.

Lemma 2. Given the DfE effort level e, the environmental impact in the competitive case is

(1) If $\frac{1-e\beta_1}{1-e\beta_2}\leq\frac{2c_r}{\phi c_n}-\frac{1}{c_n(1-e\beta_2)}$, then $E^{M*}=\frac{(1-\rho_1e)\delta_n(1-c_n(1-e\beta_1))}{2}$.

(2) If $\frac{2c_r}{\phi c_n}-\frac{1}{c_n(1-e\beta_2)}\leq\frac{1-e\beta_1}{1-e\beta_2}\leq\frac{(2+\phi)c_r}{3\phi c_n}+\frac{1-\phi}{3c_n(1-e\beta_2)}$, then

$$E^{M*}=\frac{(1-\rho_1e)\delta_n(2-\phi-2c_n(1-e\beta_1)+c_r(1-e\beta_2))}{4-\phi}+\frac{(1-\rho_2e)\delta_r(\phi+\phi c_n(1-e\beta_1)-2c_r(1-e\beta_2))}{(4-\phi)\phi}.$$

(3) If $\frac{1-e\beta_1}{1-e\beta_2}\geq\frac{(2+\phi)c_r}{3\phi c_n}+\frac{1-\phi}{3c_n(1-e\beta_2)}$, then $E^{M*}=\frac{((1-\rho_1e)\delta_n+(1-\rho_2e)\delta_r)(1-c_n(1-e\beta_1))}{2+\phi}$.

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